

Leveraging spatial residual attention and temporal Markov networks for video action understanding

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ABSTRACT

The effective use of temporal relationships while extracting fertile spatial features is the key to video action understanding. Video action understanding is a challenging visual task because it generally necessitates not only the features of individual key frames but also the contextual understanding of the entire video and the relationships among key frames. Temporal relationships pose a challenge to video action understanding. However, existing 3D convolutional neural network approaches are limited, with a great deal of redundant spatial and temporal information. In this paper, we present a novel two-stream approach that incorporates Spatial Residual Attention and Temporal Markov (SRATM) to learn complementary features to achieve stronger video action understanding performance. Specifically, the proposed SRATM consists of spatial residual attention and temporal Markov. Firstly, the spatial residual attention network captures effective spatial feature representation. Further, the temporal Markov network enhances the model by learning the temporal relationships via conducting probabilistic logic calculation among frames in a video. Finally, we conduct extensive experiments on four video action datasets, namely, Something-Something-V1, Something-Something-V2, Diving48, and Mini-Kinetics, show that the proposed SRATM method achieves competitive results.

1. Introduction

The revival of convolutional neural networks (CNNs) has led to significant advances in computer vision and is currently advancing state-of-the-art techniques for video action understanding and action recognition as well (Li et al., 2020; Sudhakaran, Escalera, & Lanz, 2020; Wang & Cherian, 2021; Yang, Xu, Shi, Dai, & Zhou, 2020). Nevertheless, current solutions for video action understanding tasks are still long away from being practically applicable arguably due to the property of video data and the complexity of human actions in real world. Video action understanding is an important component of machine learning and computer vision in the fields of surveillance, security, patient monitoring, and sports video analysis (Luo & Yuille, 2019; Sudhakaran et al., 2020).

What is required to understand a video? We humans need to partially memorize the video context and watch the complete video over time. Furthermore, according to chronological video clips, we can understand the content of the video presentation correctly (Wei et al., 2021). As shown in Fig. 1, we can see the seven frames from start

to end: the human hand pushes the faucet from right to left. Thus, spatio-temporal information must be considered in video action understanding. With deep learning being extensively adopted in the field of computer vision, deep 2D/3D CNNs for video action understanding are engineered to learn better features in the joint spatio-temporal feature space. The video action understanding architecture is dominated by two main types of frameworks, namely, 3D CNN (Liu, Tang, Li, & Rehg, 2020) and two-stream (Feichtenhofer, Fan, Malik, & He, 2019; Gong et al., 2021; Peng, Zhao, & Zhang, 2019). The 2D CNN (Hussein, Gavves, & Smeulders, 2019) does not effectively model the temporal dimension of the video and cannot consider temporal relationships. Subsequently, 3D CNN (Tran, Bourdev, Fergus, Torresani, & Paluri, 2015) is designed to capture the temporal features of the video. The 2D network extracts image-level features on which temporal aggregation is performed, and the 3D CNN learns spatio-temporal features directly from the video clips.

Temporal relationships pose a challenge to video action understanding. Temporal redundant information can be exploited by while

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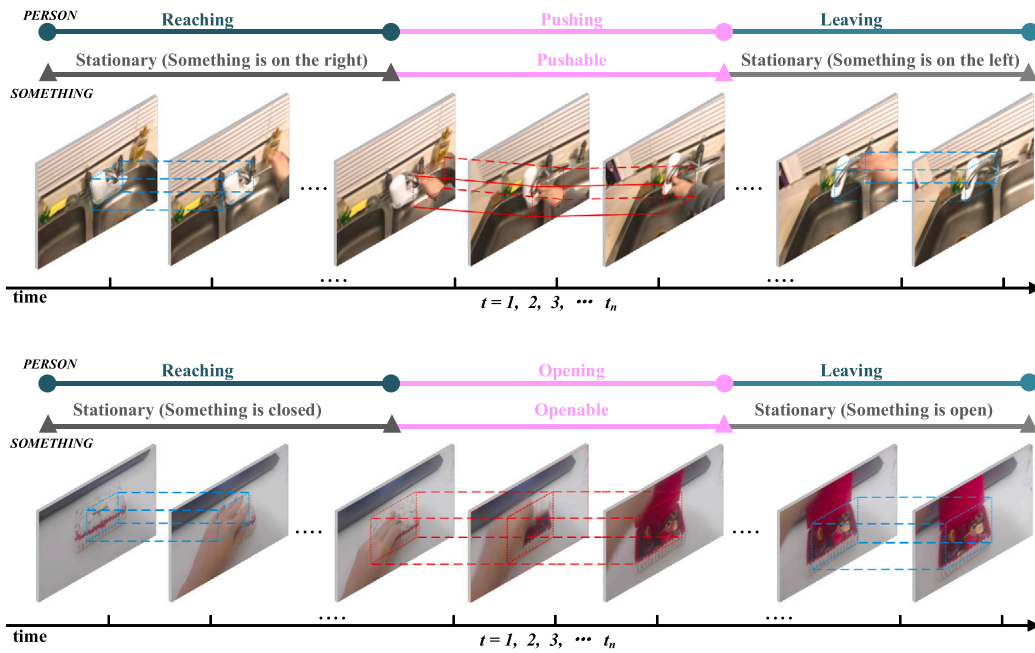


Fig. 1. Examples show multiple instances of different actions. Our model achieves promising recognition performance on these variety of actions. The person (circle) and object (triangle) entities are separated from the video and both have independent timelines (upper two rows). The segments in pink (pushing, pushable; opening, openable) represent the action sequence. A specific activity requires that the action and the interaction object appear in parallel, and nn action will trigger a result. The first line of images is labeled: “Pushing something from right to left”; The second line of images is labeled: “Opening something”.

obtaining better results. Therefore, the effective use of temporal information while extracting significant spatial features is a key issue in video action understanding. 2D CNN-based methods are two-stream inflation of 3D ConvNet (Qiu, Yao, & Mei, 2017), which is based on the expansion of 2D ConvNet: the filters and pooling kernels of very deep image classification ConvNets are extended to 3D, thereby making the learning of seamless spatio-temporal feature extractors from a video possible. Two-stream (Feichtenhofer et al., 2019; Gong et al., 2021) methods find that different streams are responsible for various tasks; one stream is responsible for presenting spatial features, and the other stream is responsible for capturing temporal features. The different features of the two streams are fused, and video action recognition is performed on the basis of fused features (Yu, Wang, Zhang, & Ren, 2020).

Although CNNs are extensively adopted in video action understanding, these 2D and 3D still suffer from some inherent deficiencies. For example, 3D CNNs use a video clip and output a score for each action category via multi-layer calculations without compelling evidence regarding the actions in a video. Recently, several studies (Ali & Bouguila, 2019; Jin, Liu, Li, He, & Zhang, 2019; Koller, Camgöz, Ney, & Bowden, 2020) stated that Markov could capture temporal information in different video tasks. Liu, Wang et al. (2018) illustrated the inference process of the two coupled dynamic Markov networks to obtain the joint video object discovery and segmentation. More specifically, object discovery is conducted through the object proposal graph that represents the correlations of object proposals among multiple frames, which are built under the help of the spatio-temporal object segmentation tube obtained by object segmentation on the image. Capable and reliable temporal modeling capabilities are the keys to improving the video action understanding performance. However, they still have some gaps, and the problem (i.e., how to effectively and efficiently model temporal structures with significant variability and complexity) is far from being solved.

In this paper, we propose a novel two-stream network: leveraging spatial residual attention and temporal Markov (SRATM) for video action understanding. In a nutshell, the SRATM model incorporates spatial and temporal networks. Spatial residual attention consists of

Conv-hybrider and Variant branches. Conv-hybrider can gradually refine the feature maps via stacking different conv blocks. The different scale residual features are between adjacent pooling layer in the variant branch to collect global information with original feature maps with the whole image. Different types of attention can be captured in different branches in Spatial residual attention. The temporal stream is constructed by temporal Markov networks, and it is trained to model temporal information. The temporal context of a stack of video frames is placed into the trained Markov networks. Eventually, we apply a fusion layer to combine spatio-temporal features to determine video action understanding.

The main contributions of this work as follows:

- (1) We present a novel two-stream model SRATM, which leverages spatial feature and temporal relationships to facilitate understanding action videos. The residual attention module being designed is constructed by Conv-hybrider and Variant branches. We devise a residual attention module to generate semantically strong features and then learn effective spatial feature representation in a video.
- (2) The temporal Markov network is leveraged to encode the temporal relationships of the previous frames in a video for better action reasoning since the weighted logical rules and probabilistic graphical models. The spatial and temporal features are fused together, and then the spatio-temporal information is used to significantly improve the recognition performance for video actions that require temporal reasoning.
- (3) We evaluate the SRATM on multiple challenging benchmarks, including Something-Something-v1&v2, Diving48, and Mini-Kinetics. We perform an extensive experiment analysis and visualization of the proposed module to study its effectiveness in video action understanding.

2. Related work

2.1. Video action understanding with deep learning

To understand a video, we need to relate the present frames to the earlier frames, putting the observed features in context. At present,

2D/3D CNNs are the more widely used approaches in video action understanding tasks.

2D & 3D CNNs. 2D CNNs operate independently in the spatial dimension, and a single 2D network structure cannot join temporal information. Existing effective video action understanding methods (Dave, Chen, & Shah, 2022; Gao, Liu, Zhang, Li, & Qin, 2023; Ji et al., 2023; Lin, Gan, & Han, 2019; Shang, Li, Xiao, Ji, & Chua, 2021; Wang et al., 2016; Xiao et al., 2021) use 2D CNNs (Zhang & Wang, 2021) directly. The proposed TSM model facilitates information exchange between adjacent frames by shifting the channel along the time dimension of different frames. A multi-resolution CNN to capture local spatio-temporal information, (Lin et al., 2019) proposed a connectivity framework for features in the time domain. However, 2D CNNs based on individual frames do not model temporal information well. Temporal Segment Networks (TSN) (Wang et al., 2016) combines a sparse temporal sampling strategy with video-level supervision to enable efficient and effective learning using whole action videos. 3D CNNs are the 3D kernel convolution of a cube formed by stacking multiple consecutive frames on top of each other. Thus, 3D CNNs are leveraged to learn spatio-temporal features jointly. For example, Chen et al. (2021) and Wei et al. (2022) developed a direct implementation of video motion recognition using deep networks through 3D convolutional networks. Their 3D feature extractor operates in the spatial and temporal dimensions to capture motion information in the video stream, and the final feature representation combines information from all channels.

Attention mechanisms. Recently, the attention mechanism (Li, Xie, Zhang, & Shi, 2023; Liu et al., 2020; Qiu et al., 2021; Ullutan, Iftekhhar, & Manjunath, 2020; Wang et al., 2018, 2020; Yin, Shen, Guan, Zhou, & Yang, 2020) has been widely adopted because of its superior representation to enhance the performance of the model. Residual attention is a kind of CNN, that is developed using attention mechanism and incorporates with state-of-art feed-forward network architecture. Gong et al. (2021), Lan, Zhang, Zhang, and Tao (2023) and Peng et al. (2019) proposed a two-stream collaborative learning method with spatio-temporal attention to learn discriminative static and motion features for video classification.

2.2. Spatio-temporal modeling

CNNs provide pyramidal features at different levels of the hierarchy. The underlying features are less useful for extracting discriminative motion cues (Xie, Sun, Huang, Tu, & Murphy, 2018). The application of 1D convolutional layers on top of 2D CNN (Zhang & Wang, 2023) for video motion recognition was proposed in Li et al. (2020). These works showed that applying full 3D and separable 3D convolution on top of a 2D CNN is more effective in extracting spatio-temporal features.

A straightforward approach to temporal modeling is to use 3D CNNs. Zhu, Xu, Hui, Lu, and Tao (2019) used two-layer MLPs and static frame features to model temporal information. In this approach, the feature extraction of each frame is isolated, and only high-level 2D CNN features are considered for temporal relation learning. Temporal Pyramid Network (Yang et al., 2020) and ATFR (Fayyaz et al., 2021) leveraged the feature hierarchy to deal with the variance of temporal information (i.e., differences in visual tempos are used to achieve temporal modeling). The temporal offset module (TSM) (Lin et al., 2019) shifts part of the channels along the time dimension and then aligns the offset features to facilitate information exchange between neighboring frames. TSM can be inserted into a 2D CNN for temporal modeling (Chen et al., 2022). Li et al. (2020) built a TEA model to integrate the short-range motion modeling with the whole spatio-temporal feature learning approach. Multiple temporal aggregation efficiently enlarges the temporal receptive field for long temporal modeling.

2.3. Hidden Markov models

The application of Hidden Markov Models (HMMs) in face recognition started from the mid to late 1990s. HMMs have a strong ability in dynamic time series modeling (Li, Wu, Sun, & Wang, 2021) and pattern recognition (Ali & Bouguila, 2019). HMMs have achieved great success in temporal modeling, which is a hot issue in computer vision. Markov is used for modeling temporal relationships in many fields (Alaa & van der Schaar, 2018) and has improved the accuracy of model. HMMs have some unique advantages when applied to pattern recognition because of their powerful temporal modeling capability, mainly in the “accuracy” capability. Liu, Wang et al. (2018) employed coupled dynamic Markov networks to compute by combining the incoming messages from both its forward and backward neighborhood frames in a video. Ali and Bouguila (2019) and Jin, Zhu, and Mu (2022) proposed the use of HMMs for action recognition and derived the mathematical model for the variational learning of Beta-Liouville HMMs. The HMMs-based models described above are not strong state-of-the-art hybrid neural network-based or even CNN-LSTM-HMM models.

More recent methods depend on recurrent neural networks (without HMMs) for sequence modeling, with an intermediate fusion of multiple detection segment features (Wei et al., 2021). However, the reliance on frame labels hinders the application to realistic video action scenarios. Koller et al. (2020) proposed a hybrid Multi-Stream CNN-LSTM-HMMs model that was applied to video sign language action recognition, and the algorithm was endowed with the potential to learn from weak supervision to obtain strong and reliable frame labels. In addition this model is the first to investigate multi-stream hybrid HMM method using CNN techniques and LSTM technique, and proves that HMMs hybrid with CNNs is feasible.

In this paper, we propose a temporal Markov module for temporal modeling in video action understanding tasks. In Section 3.2, we derive, a strict mathematical formulation of how to model the temporal information relationship of a video using a HMM. Thus, temporal Markov module increases the interpretability of temporal relationships in videos. Here, we extend temporal Markov module to further include temporal voxels into CNN learning and inference process, which results in a model encoded with rich spatio-temporal information.

3. Method

In this section, we propose SRATM for video action understanding. The SRATM model, which consists of a backbone feature extractor, spatial residual attention network (Section 3.1), and temporal Markov network (Section 3.2), is depicted in Fig. 2. Spatial residual attention network has two branches: one is Conv-hybridizer branch and the other is Variant branch. We first describe the spatial residual attention module details and derive the Markov network for temporal modeling. Spatial and temporal streams are implemented using spatial residual attention and Markov networks, respectively; the SoftMax scores of which are combined by late fusion.

3.1. Spatial residual attention network

Our spatial residual attention network robustly removes background noise from features and obtains strong reliable feature information. By combining the residual blocks and the attention mechanism to form the module (Song et al., 2020; Wang et al., 2018), the module helps the SRATM model to improve its recognition accuracy.

In ResNet model, residual learning is represented as $y = F(x) + x$. $F(x)$ represents the residual mapping generated by CNNs. x and y are the input and output vectors, respectively. As shown in Fig. 3, Spatial Residual Attention module is divided into two branches: Conv-hybridizer and Variant branches. The Conv-hybridizer branch performs feature processing, and the variant branch can serve as feature selectors to enhance favorable features and restrain the noise of the main features. Similar to

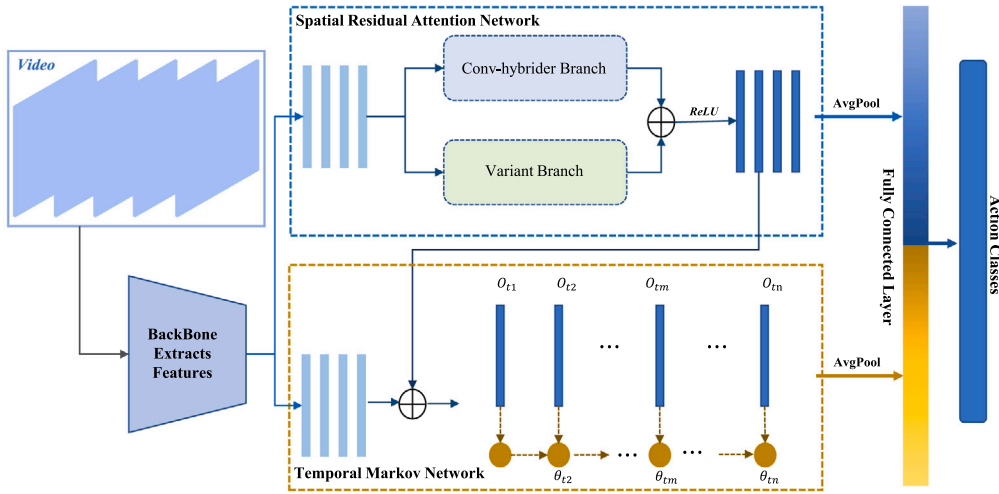


Fig. 2. The proposed SRATM model performs spatio-temporal modeling efficiently. (1) The Spatial Residual Attention network contains Conv-hybrid and Variant branches. (2) Temporal Markov network captures the temporal information via probabilistic logic calculation.

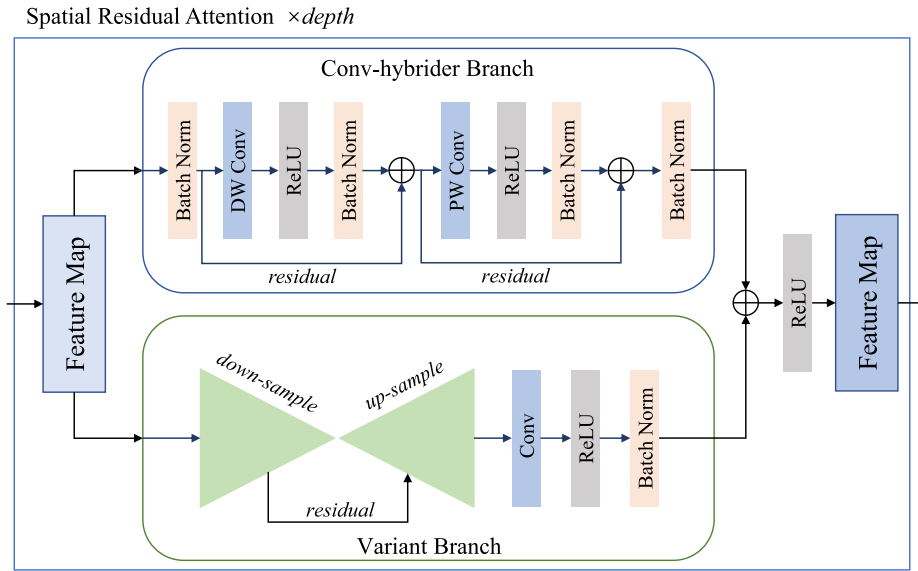


Fig. 3. Illustration of an spatial residual attention. The spatial residual attention consists of Conv-hybrid and Variant branches. Spatial residual attention module processes the input features to create the selection and localization of significant spatial features. This module captures interaction features by treating person/objects as nodes and their interactions as edges. DW indicates depthwise convolution($kernel_{size} = 3$) and PW means pointwise convolution($kernel_{size} = 1$); the 'xdepth' is the number of repetitions of the Spatial Residual Attention module.

ResNet, we also use the element-wise summation to process the output of Conv-hybrid and Variant branches in Fig. 3.

$$F_i(x) = C_i(x) + V_i(x), \quad (1)$$

where $C_i(x)$ is the last output of the Conv-hybrid branch and $V_i(x)$ is obtained after processing by the variant branch. i denotes the i th spatial position.

The Conv-hybrid branch contains depthwise convolution followed by pointwise convolution. DW and PW denote depthwise and pointwise convolutions, respectively.

$$\begin{aligned} x' &= \text{BN}(\sigma(\text{DW-Conv}(\text{BN}(x)))) + \text{BN}(x), \\ x_c &= \text{BN}(\sigma(\text{PW-Conv}(x))) + x', \end{aligned} \quad (2)$$

where the σ is the ReLU activation and the BN is the batch norm operation.

The variant branch consists of down-sampling, up-sampling, convolutional structure, and activation function components. The input

features are subjected to down-sampling (max pooling) several times to increase the receptive field rapidly, after which a small number of remaining units are extracted. After performing the down-sample operation to reach the lowest resolution, the global information is expanded by a symmetric top-down architecture. The processing of the spatial feature map is mainly preceded by down-sampling and up-sampling processes, which are used to obtain the global features of the spatial features. Then, the global high-dimensional features extracted by up-sampling are combined with the previous residual features to obtain rich features. It can be formulated as:

$$\begin{aligned} x'' &= U_s(D_s(x) + x), \\ x_v &= \text{BN}(\sigma(\text{Conv}(x''))), \end{aligned} \quad (3)$$

where $D_s(\cdot)$, $U_s(\cdot)$, and $\text{Conv}(\cdot)$ denote down-sample, up-sample, and convolution operations, respectively. σ is an activation operation. $V(x)$ is obtained after convolution layers and *sigmoid* activation function in the Variant branch. The *sigmoid* activation layer normalizes the output values.

Finally, we can obtain $F_i(x) = \text{ReLU}(x_c + x_v)$ in Fig. 3. We add the ReLU activation function in the Variant branch to end up with a nonlinear feature vector, thereby making the optimization of the residual mapping easier than the original mapping (He, Zhang, Ren, & Sun, 2016).

Residual attention learning not only maintains the good properties of the original features but also provides them the ability to bypass variant branch and forward features to the top layer while suppressing the feature selection ability of the Variant branch. Spatial Residual Attention network works as feature attention that enhances better and more fertile features.

3.2. Temporal Markov network

We conduct the probabilistic logic calculation procedure for video temporal information by HMMs. Many works perform temporal feature analysis by first cutting the video into many frames and extracting features (mainly temporal information) for each frame ($T' \in T$), thereby obtaining a sequence of feature vectors $O = \{O_{t_1}, O_{t_2}, \dots, O_{t_n}\}$ ($O \in T'$ is an input video segment and t_n represents the end time). Each feature vector is a random vector that satisfies a certain probability distribution, and this sequence of random vectors varies over time.

We define the hidden Markov model θ with a set of parameters (N, A, B, π) to be the temporal hidden Markov model, which can be expressed as $\theta(N, A, B, \pi)$, where N is the number of states, A is the state transition probability matrix, B is the observation probability matrix, and π is the state initial probability vector distribution.

According to Bayes' principle, θ has posterior probability:

$P(O|\theta) = P(O|\theta) \times P(\theta) / P(O)$, where $P(\theta)$ is the prior probability of model θ and $P(O)$ is the probability of training sample O .

As the number of training samples increases, the size of $P(O|\theta)$ will increase without limit, whereas the size of $P(\theta)$ is finite and constant. Therefore, θ has a high prior probability as long as a sufficient number of training samples is guaranteed.

$$\begin{aligned} \theta_m &= \arg \max_{\theta} (P(O|\theta)) \\ &= \arg \max_{\theta} (P(O|\theta) \times P(\theta)), \end{aligned} \quad (4)$$

where θ_m is the optimal model we seek.

The recognition process of temporal features can be described as

$$\begin{aligned} r &= \arg \max_{1 \leq r \leq L} (P(\theta_r|O)) = \arg \max_{1 \leq r \leq L} (P(O|\theta_r) \times P(\theta_r)) \\ &\text{subject to } \sum_{r=1}^L P(\theta_r) = 1, \end{aligned} \quad (5)$$

where r is the recognition result, and L is the total number of pattern categories. In this decision formula $P(\theta_r)$ is the prior probability of pattern r .

We use the prior probability $P(\theta)$ expression to find the higher prior probability in the model space (N, A, B, π) . According to information theory, the entropy of the θ probability distribution:

$$H(\theta) = - \sum_{\theta} P(\theta) \log P(\theta) = - \int P(\theta) \log P(\theta) d\theta. \quad (6)$$

According to the famous maximum entropy principle, the prior probability of θ is:

$$P(\theta) = \exp(-\lambda H(\theta)) / \alpha, \quad (7)$$

where α is a normalized constant, and λ is usually set to 1.

The model with maximum posterior probability for a training sample O with a fixed number of states N is

$$\theta^* = \arg \max_{\theta(A, B, \pi)} (P(O|\theta) P(\theta)). \quad (8)$$

According to Eqs. (6) and (7):

$$P(O|\theta) P(\theta) = \log(P(O|\theta)) - \lambda H(\theta). \quad (9)$$

This means that we need to find the minimum value of the following objective function E :

$$E(\theta) = -\log(P(O|\theta)) + \lambda H(\theta). \quad (10)$$

Then the maximum posterior model θ^* needs to satisfy:

$$\theta^* = \arg \min_{\theta(A, B, \pi)} E(\theta). \quad (11)$$

The observed multi-feature sequence in a video $O = \{O_{t_1}, O_{t_2}, \dots, O_{t_n}\}$ is used as an input, training model θ .

We propose the Temporal Hidden Markov module, which has a pending parameter set (N, A, B, π) , using a shrinkage training method with maximum posterior probability as the optimization criterion. Such a temporal model is designed to make the observed sequential state features in real video temporal features, which are more consistent with the structure of temporal features and can extract more temporal information.

3.3. Network architecture

Inspired by Liu, Li, Luo, Loy, and Tang (2018) and Peng et al. (2019), the SoftMax function is leveraged. Fusion is performed so that each element of the vector takes values between (0, 1). We obtain the overall objective for a backbone network with our proposed SRATM, thus:

$$L_{total} = L_{SRA} + \kappa \theta^*, \quad (12)$$

where L_{SRA} and θ^* are the original Cross-Entropy loss for Spatial Residual Attention module and Temporal Markov module loss, respectively. κ is a balancing coefficient.

4. Experiment

We first provide the datasets, implementation details, and comparison to the state-of-the-art methods. Then, we carry out ablation studies and analyze the impact and performance of each SRATM component. Extensive experiments are conducted to evaluate the SRATM model against plenty of state-of-the-art video understanding models on Something-Something-V1, Something-Something-V2, Diving48, and Mini-Kinetics benchmarks.

4.1. Datasets and implementation details

Datasets. We employ three widely-used and challenging benchmarks, i.e. Something-Something-V1, Something-Something-V2 (Goyal et al., 2017), and Diving48 (Li, Li, & Vasconcelos, 2018). The number of videos in the Something-Something-V1&V2 datasets is 108,499 and 220,847, respectively. The two datasets are large collections of labeled video clips that show humans performing pre-defined basic actions on everyday objects, and they both have 174 classes. Machine learning models provide a fine-grained understanding of the basic actions that appear in the videos. Diving48 is a diving competition video dataset that consists of 18000 video clips of 48 unambiguous diving sequences. Diving48 is characterized by an action recognition dataset with no significant bias for static or short-term motion representations, and thus can be employed to test the ability of different models in capturing long-term dynamic information. Mini-Kinetics (Xie, Sun, Huang, Tu, & Murphy, 2017) is a smaller dataset compared to the full Kinetics-400 (Kay et al., 2017) dataset and consists of 200 categories. Mini-Kinetics consists of 200 categories with the most training examples. For each category, it randomly samples 400 examples from the training set, and 25 examples from the validation set, resulting in 80,000 training examples and 5,000 validation examples in total.

Implementation details. Our model is implemented with Pytorch. We implement BN-Inception and ResNet-50 as the CNN backbone. The model is initialized with ImageNet pre-trained weights. The $\times depth$ of

Table 1

Demographic performance comparison with competitive methods was measured by top-1 and top-5 accuracy(%) on Something-Something-V1 and V2. “Improve” denotes the performance improvement of SRATM. MN2 refers to MobileNet-V2. The best and the second-best results are highlighted in boldface and underlined, respectively. “Improve” represents the percentage improvement of the best model to the second-best model.

Method	Backbone	Frame	Something-V1		Something-V2	
			top-1	top-5	top-1	top-5
TSN (Wang et al., 2016)	BN-Inception	16	19.50	47.3	30.0	60.5
ECO (Zolfaghari, Singh, & Brox, 2018)	BNInc+ResNet	16	41.40	–	–	–
R(2+1)D (Tran et al., 2018)	ResNet-34	32	45.70	–	–	–
S3D-G (Xie et al., 2018)	InceptionV1	64	48.20	78.7	–	–
ABM-c (Zhu et al., 2019)	ResNet-50	32 × 3	50.09	–	63.9	–
Martinez et al. Martínez, Modolo, Xiong, and Tighe (2019)	ResNet-50	NA	50.10	–	–	–
GSM (Sudhakaran et al., 2020)	InceptionV3	32	51.68	–	–	–
TRN MultiScale (Zhou, Andonian, Oliva, & Torralba, 2018)	BN-Inception	8	34.44	–	48.8	77.6
TRN Two-stream (Zhou et al., 2018)	BN-Inception	8+8	42.00	–	55.5	83.1
TSM (Lin et al., 2019)	ResNet-50	16	44.80	74.5	58.7	84.8
TSM Two-Stream (Lin et al., 2019)	ResNet-50	16+8	49.60	79.0	63.5	<u>88.6</u>
AdaFocusV2-TSM (Wang et al., 2022)	MN2+ResNet-50	8+12	49.60	–	61.3	–
ATFR (Fayyaz et al., 2021)	ResNet-50	32	–	–	61.8	87.9
TDN (Wang, Tong, Ji, & Wu, 2021)	ResNet-50	8	<u>52.3</u>	–	<u>64.0</u>	–
AGPN (Chen, Ge, Liu, Cai, & Sun, 2023)	ResNet-50	8	51.6	<u>80.9</u>	63.1	<u>88.6</u>
VidTr (Zhang et al., 2021)	ViT-B	16	–	–	61.9	–
VidTr (Zhang et al., 2021)	ViT-L	32	–	–	63.0	–
TimeSformer (Bertasius, Wang, & Torresani, 2021)	ViT-B	8	–	–	59.5	–
TimeSformer (Bertasius et al., 2021)	ViT-L	64	–	–	62.4	–
SRATM	ResNet-50	8	52.1	81.3	64.8	89.1
SRATM	InceptionV3	32	52.53	81.98	64.51	89.68
Improve	–	–	0.44%	1.33%	1.25%	1.22%

the spatial residual attention are set to 3. Dropout is set to 0.5 and applied to the Something-Something-(V1&V2) and Diving48 datasets. The input dimension of InceptionV3 is 224×224 . The input dimension of InceptionV3 reduces the computational complexity without reducing in performance. We train the model with Stochastic Gradient Descent (SGD) with 0momentum 0.9 and weight decay of $1e-4$.

On the Something-Something-(V1&V2) datasets, we begin with a learning rate of 0.01 and the epoch of the experiment is set to 100. We report top-1 and top-5 accuracy (%). For Diving48 dataset, we start with a learning rate of 0.01, and stop at 160 epochs. We conduct experiments with setting different batch sizes (2/64) and trained the model on 8 GPUs.

4.2. Comparison to the state-of-the-art methods

Results on Something-Something-(V1&V2). Under the different metrics, Table 1 shows the performance of various competitive algorithms. Something-Something-(V1&V2) are challenging datasets, and the video has just one activity with clear start and end. Thus, the temporal relation between frames is shown more logically. Video action understanding cannot rely on a single frame to identify the video content, but to rely on the continuous frame (e.g., video labels in Fig. 1: “Pushing something from right to left”; “Opening something”). We compared SRATM with current state-of-the-art methods in Tables 1 and 2. The TRN and TSN models developed using 2D CNN is designed to learn and reason about the temporal dependencies between video frames at multiple time scales. Although TSM captures the temporal dependency of frames, the performance is still lower than the state-of-the-art methods, thereby showing the limitations of 2D CNN in time modeling and the important temporal alignment of videos. However, the performance of TRN, TSN and TSM is not the best, which reveals the significance of temporal relationship and feature fusion. ECO, S3D, and Martinez all employ 3D CNN in video tasks, but they did not achieve competitive accuracy. The likely reason is that the models do not model temporal sufficiently well.

In Table 1, the SRATM results show the top-5 accuracy is improved by 1.33% on Something-Something-V1 dataset compared with the second-best model. TSM and TRN Two-Stream models achieve better accuracy on different datasets compared with their non-Two-stream architectures, and these experiment results can be inferred

that the Two-Stream network is strongly robust. The proposed SRATM using ResNet-50 compares with the TDN (Wang, Tong et al., 2021) and AGPN (Chen et al., 2023) and shows strong performance on all the metrics on two datasets. Specifically, our SRATM using ResNet-50 achieves +0.5 higher accuracy (89.1% v.s. 88.6%) than the competitive AGPN (Chen et al., 2023) on Something-Something-V2. We also compare our approach to AGPN (Chen et al., 2023) (Ours: 25.3M v.s. AGPN: 27.6M) in term of model parameters to show our model with less complexity.

In addition, we provide the Transformer-based methods, namely TimeSformer and VidTr, as the comparison methods. As shown in Table 1, TimeSformer (Bertasius et al., 2021) with ViT-L (Dosovitskiy et al., 2020) backbone achieves a top-1 performance of 62.4%. VidTr (Zhang et al., 2021) using ViT-L achieves a top-1 performance of 63.0%. Our SRATM obtains slightly better performance than the original TimeSformer with ViT-L. The small performance improvement might be ascribed to the fact that the CNN-based SRATM uses spatial and temporal streams to learn the action region features better. Whether based on CNN or Transformer as the backbone, in the case of little difference in the scale of the backbone, it effectively combines spatial and temporal information to learn better features for action modeling. Spatial and temporal streams are combined for better learning with rich spatio-temporal information. The proposed SRATM is able to make the correct prediction and the attention showed that the SRATM is able to focus on the action-related regions across time (See Fig. 4). All these results further indicate that our SRATM model can boost video action understanding accuracy effectively.

Results on Diving48. The diving movements in the Diving48 dataset are a coherent set of movements and therefore require the modeling of long-term temporal dynamics. In Table 2, our proposed SRATM method is superior to the recent methods. Our model can cope with long-term temporal modeling and remove spatial background noise from the video effectively, which allows the model to keep the focus on the athletes’ actions. SRATM gains a 2.83% performance improvement compared with other methods. The results show the efficacy of our proposed method. It can also be observed from (e) and (f) in Fig. 4 that our model has the capacity to capture the actions of the athlete in the video for a long time and in focus.

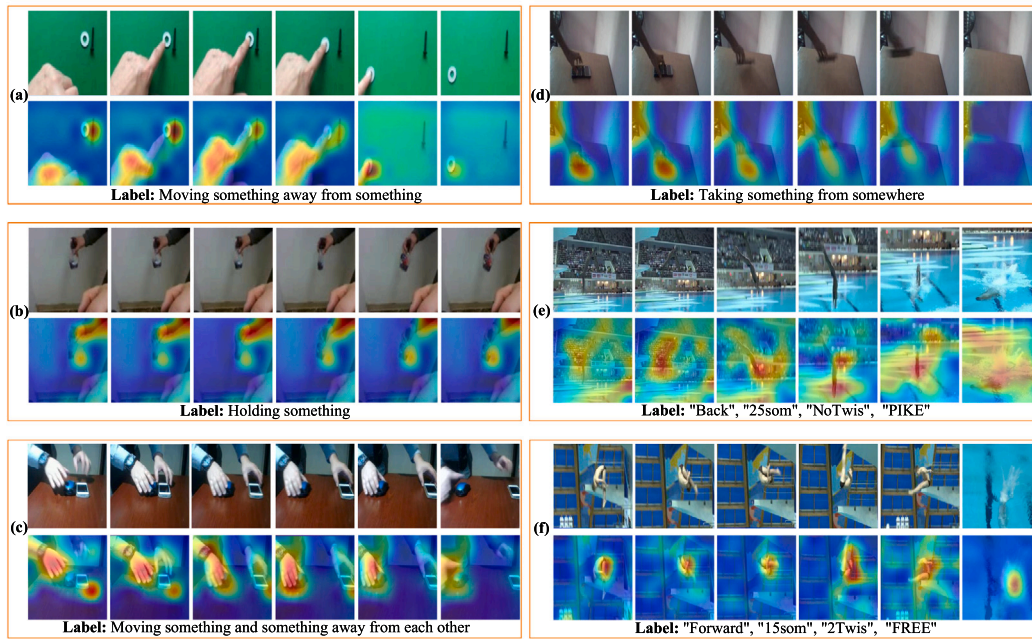


Fig. 4. Visualization for favorable features extracted by our model on Something-Something-(V1&V2) and Diving48 datasets. We randomly take six frames as the input to visualize and show six examples. (a), (b), (c) and (d) examples are from the Something-Something-(V1&V2). The proposed SRATM focuses on the action more accurately from beginning to end in (a)–(d). (e) and (f) examples are from Diving48. It can be noticed that the distribution of attention across time and our method can effectively capture more fine-grained action information in the diving video.

Table 2

Comparison to state-of-the-art on Diving48. “Improve” represents the percentage improvement of the best model to the second-best model. The best result is highlighted and the second-best is underlined. “Improve” represents the percentage improvement of the best model to the second-best model.

Method	Pre-training	Accuracy(%)
TSN (Wang et al., 2016)	ImageNet	16.77
TRN (Zhou et al., 2018)	ImageNet	22.80
R(2+1)D (Zhou et al., 2018)	Kinetics	28.90
P3D (Qiu et al., 2017)	ImageNet	32.40
C3D (Tran et al., 2015)	ImageNet	34.50
Kanojia (Martínez et al., 2019)	ImageNet	35.64
GST (Luo & Yuille, 2019)	ImageNet	<u>38.80</u>
SRATM	ImageNet	39.90
Improve	–	2.83%

Table 3

Comparison to the state-of-the-art on Mini-Kinetics. Accuracy measured by Top-1 classification accuracy (%).

Method	Backbone	Accuracy (%)
TSN (Wang et al., 2016)	ResNet-50	72.1
AR-Net (Meng et al., 2020)	ResNet-50	71.7
AdaFocus (Wang, Chen et al., 2021)	ResNet-50	72.2
AdaFuse (Meng et al., 2021)	ResNet-50	72.3
DKP (Kim, Jain, Lee, Yun, & Porikli, 2021)	ResNet-50	72.7
SRATM	ResNet-50	73.7

Results on Mini-Kinetics. To show the flexibility of the proposed SRATM, we experiment on Mini-Kinetics (Xie et al., 2017) with ResNet-50 backbone. We report the performance of top-1 classification accuracy. As shown in Table 3, we again demonstrate the flexibility of our SRATM. Specifically, with ResNet-50, our trained model reaches 73.7% in Top-1 accuracy. For Mini-Kinetics, our SRATM achieves +1.0 (Our 73.7% vs DKP 72.7%) improvement in Top-1 accuracy using ResNet-50 as backbone compared with previous methods like DKP (Kim et al., 2021). The experiment results show that the proposed SRATM significantly outperforms all compared methods. This indicates that our method is beneficial to recognize the actions in videos.

4.3. Ablation analysis

To evaluate the contribution of each component of our method, we conducted extensive ablation experiments on the Something-Something-V1 validation dataset. As shown in Table 4, we ablate SRATM to understand the contribution of each design and then report the experiment results.

Spatial residual attention To understand the benefits of the spatial residual attention module, we compare the effect of the different branches in SRA. As shown in Table 4, we show the experiment results for different combinations of components (*i.e.*, Baseline+Conv-hybrider Branch and Baseline+Variant Branch). We compare three different configurations in the SRA module. We find that the sum of the values of the outputs of the Conv-hybrider branch and Variant branch can keep useful information and complete the selection and localization of significant features. In fact, the SRA using Conv-hybrider and Variant branches achieves better accuracy. Note that experiments on the module with the residual attention show higher accuracy, and it can be inferred that it can learn more representative features. We leverage residual attention module to select the valuable spatial features based on the fusion of the residual features. We can see that the baseline+SRA achieves better results compared with results from the different components, showing the effectiveness of spatial modeling. Residual attention learning can employ the same feature map to reduce the loss of effective features and thus enhance feature abundance. By combining both modules into a baseline network, our highest performance can reach 52.5% in Top-1 accuracy, as shown in Tables 1 and 4. In summary of the above, we are able to conclude that the spatial residual attention captures essential spatial information. Moreover, the fertile output of spatial residual attention provides an indispensable contribution to the overall model.

Temporal Markov Network. To account for the significance of temporal modeling for video action understanding, we present experiment results for different components in Table 4. Specifically, we compare different combination settings on the Something-Something-V1 dataset. As shown in Table 4, the precision of SRATM is 14.3% higher than that of Baseline-SRA. We can observe that the performance

Table 4

Ablation study to on Something-Something-V1. Baseline-SRA: spatial feature learning without temporal information. “Improve” represents the percentage improvement of the best model to the second-best model. Baseline+TM: the model learning without spatial residual attention.

Model	Accuracy (%)	Gain (v.s. Baseline)
BN-Inception (Baseline)	22.8	0
Baseline+Conv-hybrider Branch	44.9	+22.1
Baseline+Variant Branch	45.1	+22.3
Baseline-SRA	45.9	+23.1
Baseline+TM	41.6	+18.8
SRATM	52.5	+29.7

of SRATM is substantially improved by adding a temporal Markov network. Likewise, the experiment results for the other models in Table 1 reveal that temporal modeling must not be ignored. Activities are made up of human actions and interacting objects, and happen in the sequence of actions. The human action changes in the video, and the result of the action can be modeled very accurately by HMMs. HMMs are applied for the estimation of frame labeling. Our Temporal Markov network can model the relationship between successive frames, which improve the performance on some action categories such as “holding”, “taking”, “bending”, “pushing” and “unfolding”, because they have a causal link and are continuous in successive frames. In Figs. 1 and 4, we show multiple instances of different actions. Our Temporal Markov network does enable joint inference using temporal contextual information. The SRATM model yields 52.5% precision, whereas the Baseline-SRA only achieves 45.9% in Table 4. The experimental results of our SRATM model clearly demonstrate the benefits of HMMs for temporal action information modeling. Moreover, the previous Multi-Stream CNN-LSTM-HMMs (Koller et al., 2020) model also outshines state-of-the-art methods on a fine-grained hand action recognition dataset. Likewise, the experimental results for the other models in Table 1 reveal that temporal modeling cannot be ignored, especially for the Two-Stream model, because the Two-Stream models the temporal dimension individually. In our work, we also demonstrate that the temporal context is learned efficiently via temporal Markov. The final model achieves the best performance in terms of recognition accuracy on the three datasets. Extensive experiment results can show that the our Temporal Markov network is effective for temporal information modeling via probabilistic logic calculation.

4.4. Visualization

We demonstrate the advantages of SRATM with quantitative visualizations on Something-Something-(V1&V2) and Diving48 datasets shown in Fig. 4. We employ class activation mapping (Selvaraju et al., 2017) to highlight those image regions that show the highest activation when the sequence is classified in the last convolution layer of SRATM. We visualize six examples taken from the Something-Something-(V1&V2) and Diving48 datasets and show them in Fig. 4. The first and second rows in (a), (b), (c), (d), (e) and (f) contain image samples and class activation maps from the same video, respectively. Fig. 4 visualizes features characterized by our model for the different actions e.g., “Moving something away from something”, “holding something”, “Moving something and something away from each other”, “Taking something from somewhere” and “diving actions”. It can be noticed that SRATM can focus on recognizing people (hands) and objects (something) independently rather than reasoning about action. The proposed SRATM better describes an action by representing a feature map that covers the hand and the object, especially for the highlighted parts.

We see that the most important visual regions, such as hands and screw pads and bottles in the (a) and (b) examples, are identified, and “athletes” in the (e) and (f) examples. A careful observation of Fig. 4 shows that the video backgrounds of (e) and (f) are more

complex than those of (a), (b), (c), and (d). By our model, not only the important regions are correctly recognized but also the noise from irrelevant backgrounds is largely eliminated. Therefore, the spatial residual attention gains the benefit of reducing video background noise without significant loss of spatial information, which makes spatial features more fertile while improving the recognition of the represented features. Our SRATM model can capture various human actions in video very well in challenging situations.

5. Conclusion

In this paper, we propose the SRATM model in the video action understanding domain, including the Spatial Residual Attention network and the Temporal Markov network for spatial and temporal modeling in video action recognition task. Specifically, the Spatial Residual Attention network is composed of Conv-hybrider and Variant branches for spatial feature modeling, such a module is primarily for the model to learn more fertile spatial features. Then, for temporal modeling, the Temporal Markov network is leveraged to learn temporal information efficiently by probabilistic logic calculation. We also visualized some of the action instances to see the activation map changes for continuous video frames. Extensive experiments and ablation studies on the Something-Something-V1&V2, Diving48, and Mini-Kinetics datasets demonstrate that our SRATM outperforms the state-of-the-art methods for video action understanding.

In the future, we will investigate multi-modal deep video models in the multimedia domain to enhance the deep model’s power to understand videos under weak video level annotations. Additionally, the proposed SRATM model in this paper has some limitations, mainly in two aspects: no dynamic weighted interaction between spatial and temporal stream and secondly computational complexity.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript.

Data availability

Data will be made available on request.

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